

Final Project Description

15.465 Alphanomics

Date: Feb 17, 2025

Overview:

The final project consists of two parts. The first part requires you to conduct data analysis of candidate investment signals within a dataset that I provide to you. Your findings from Part 1 should then be treated as “ground truth” about your firm’s belief about the relationship between firms’ characteristics and expected returns. The second part is informed by your Part 1 findings and asks you to generate an investment recommendation by calculating and combining signals that you manually calculate. Ultimately, you will be asked to provide a forced investment recommendation on three stocks: exactly one stock that you recommend buying, exactly one that you recommend selling, and exactly one for which you are neutral.

Please make sure to read the instructions in detail. After reading the instructions, feel free to email the TA’s with questions.

Pre-Registration:

Before beginning the project, you must pre-register your group’s participants and specify three companies that you will analyze. These companies must be currently publicly traded, in the same business sector, and listed on stock exchanges within the same country (e.g., studying firms on Nasdaq and NYSE are acceptable because they are all listed in the U.S.). Additionally, you are not permitted to select any of the companies you analyzed for the value-investing assignment. Outside of those requirements, you are free to choose any companies you like. However, after pre-registering these three companies, you are not permitted to change them. Once you pre-register your companies, you will be assigned a specific simulated dataset to analyze.

Part 1: Data Analysis

In Part 1, you will explore Fama-MacBeth regressions, a popular method for analyzing time-series cross-sectional data. As discussed in class, Fama-MacBeth regressions are a two-step procedure used in empirical finance to analyze how different characteristics of stocks (like profitability, value, or momentum) affect their expected returns while accounting for cross-sectional correlation in stock returns. By first running cross-sectional regressions each month, then averaging the monthly coefficients over time, the method helps identify which factors have consistently predicted returns.

You can use any programming language you prefer.

<< Dataset Description >>

You have been assigned a specific simulated dataset containing information for 200 firms over 60 calendar months. You must use the assigned dataset as provided without modification. The unit of analysis is at the firm-month level. The dataset includes the following variables:

****Dependent Variable (referred to as returns):****

- Monthly stock returns (not standardized), denoted as 'ret' (a value of 'ret'=0.01 → 1%)

****Independent Variables (referred to as signals):****

(All signal variables are standardized within each calendar month to have mean=0 and standard deviation=1), such that values of each signal are expressed in term of standard deviations from the mean.

- 1) Insider Selling (denoted 'inssell') – The amount of shares sold minus the amount of shares purchased by C-Suite executives in the past three months, divided by the firm's total shares outstanding. For this assignment, focus on the CEO (sometimes called President or Managing Director), CFO (also known as Finance Director or VP of Finance), COO (occasionally titled Executive VP of Operations), and Board Directors (sometimes referred to as Non-Executive Directors or Supervisory Board Members). Zero if missing.
- 2) Book-to-Market (denoted 'booktomarket') - The ratio calculated by dividing a company's book value of equity (total assets minus total liabilities) from the most recent fiscal quarter by its current market capitalization.
- 3) Momentum (denoted 'momentum') - A stock's cumulative market-adjusted return measured over the past 12 months, excluding the most recent month to avoid short-term reversals.
- 4) Gross Profitability (denoted 'grossprofit') - A firm's gross profits (revenue minus cost of goods sold) scaled by total book assets from the most recent fiscal year.
- 5) Earnings Surprise (denoted 'earnssurprise') - The difference between actual quarterly earnings and analyst consensus forecasts at the firm's most recent quarterly earnings

announcement, divided by the market price per share of the stock five days prior to the announcement date.

- 6) SUE (denoted 'sue') - The firm's most recent quarterly earnings minus the quarterly earnings from the same quarter in the previous year, divided by the market price per share of the stock five days prior to the announcement date.
- 7) F-Score (denoted 'fscore') - The composite score ranging from 0-9 based on nine binary financial signals including profitability, leverage, and operating efficiency metrics.
- 8) External Financing (denoted 'xfin') - The net amount of capital raised through debt and equity issuance minus debt repayment and equity repurchases as reported in the firms' most recent fiscal year cash flow statement, scaled by total assets.
- 9) Volatility (denoted 'volatility')- The standard deviation of daily stock returns over the past 12 months.
- 10) Accruals (denoted 'accruals') - The difference between total net income and cash flow from operations, scaled by total assets, from the most recent fiscal quarter.

<< Required Analysis >>

Beginning note: All signals above are measured prior to the month in which returns ('ret') are measured so there is no need to create lags of the variables. You can simply regress returns on the signals' values as provided in the dataset.

Step A: Univariate Analysis

1. For each signal in the dataset, iteratively run separate monthly cross-sectional regressions of the returns variable ('ret') on a given signal and save the coefficients for you to use them later.
2. With the coefficients you saved, create time-series plots of the monthly univariate coefficients for each signal. Report all 10 plots as a 5 by 2 grid (i.e., 5 rows of 2 plots side-by-side) with legible labels for the chart axes.
3. Create a table that contains:
 - Time-series average coefficients
 - T-statistics
 - Average R-squared
 - Number of observations used
4. Create a table showing the fraction of months each coefficient is positive, and discuss whether this impacts your investment decisions

Step B: Multivariate Analysis

1. Run monthly cross-sectional regressions of returns including all signals in the regression simultaneously and save the coefficients.
2. Create time-series plots of the monthly coefficients for each signal. Report all 10 plots as a 5 by 2 grid (i.e., 5 rows of 2 plots side-by-side) with legible labels for the chart axes.
3. Create a table that contains:
 - Time-series average coefficients
 - T-statistics
 - Average R-squared
 - Number of observations used

Part C: Sharpe Ratio Analysis

1. Calculate implied Sharpe ratios from the multivariate t-statistics
2. Create a bar chart showing the implied Sharpe ratios of each signal in descending order (i.e., the first bar should correspond to the highest implied Sharpe ratio, followed by the second highest, etc.)

<< Required Written Analysis >>

Answer the following questions in detail:

1. Univariate Analysis:

- Which signals appear to have the strongest univariate relationship with returns?
- How stable are the coefficient estimates over time? Are there any notable patterns or trends in the coefficients?
- For any given signal that significantly predicts returns, what do you view as the most likely explanation for why the signal 'works'? These could be a type of behavioral bias, market friction, or risk factor as discussed in class.

2. Multivariate Analysis:

- Which variables remain significant predictors of returns in the multivariate tests?
- How do the coefficients and t-stats change, if at all, when controlling for other signals?
- What does the average R-squared tell us about the model's explanatory power?
- Interpret the coefficient magnitudes in economic terms. Provide a plain English description of what each coefficient value in the table you produce implies for variation in monthly stock returns [Hint: recall that the signals are given to you as standardized values but the returns are not].

3. Sharpe Ratio Analysis:

- Which four signals appear to offer the strongest performance in terms of Sharpe ratios (either positive or negative)?
- How might these results inform investment strategy design?

<< Technical Notes >>

- For calculating standard errors, use the Fama-MacBeth approach described in the notes: $\text{Standard Error} = \text{std}(\text{monthly coefficients}) / \sqrt{\text{number of months}}$
- $T\text{-statistic} = \text{average coefficient} / \text{standard error}$
- Judge a signal as being statistically significant if the $|t\text{-statistic}| > 1.645$
- Report all coefficients in decimal form (e.g., 0.002 rather than 0.2%)

Part 2: Research and Recommendation

Building on your analysis from Part 1, you will now apply your empirical findings to make real-world investment recommendations. Your Part 1 results should be treated as “ground truth” about the relationship between firms’ and expected returns (i.e., the define your firm’s beliefs about which and to what extent the signals predict returns). This means that if a signal positively predicted returns in your Part 1 analysis, you should assume it will continue to positively predict returns. Similarly, if a signal negatively predicted returns or was statistically insignificant, you should maintain those expectations.

<< Data and Signal Requirements >>

For your three pre-registered companies, you will collect the most up-to-date values of all signals that were statistically significant at the 10% level ($|t\text{-statistic}| > 1.645$) in your Part 1 multivariate analysis. The signals should be calculated following the same definitions used in Part 1 using the mostly recent available values as of the time of this project (feel free to use end of April as the cutoff date)

<< Required Analysis >>

You are expected to write a **professional report** that explains and discusses what you did and the results you obtained for the following steps:

Step A: Signal Collection and Measurement

1. For each significant signal from Part 1:
 - Calculate the most recent available value for each of your three companies
 - I. Part of the assignment is for you to research where the signals can be sourced. You are required to source all financial information from companies’ official filings with the SEC. For price and return data, please use a reputable provider such as Bloomberg or Yahoo! Finance.
 - II. Create an easy-to-read table that contains for each company and each signal the following variables:
 1. The value of signal
 2. The source of your signal value (naming the source is sufficient e.g., 10K for the fiscal period ending YYYYMMDD or Yahoo! Finance price charts)
 3. The ‘As-Of’ date that indicates the date to which the signal value applies (e.g., the date when price was measured, the date of the fiscal

quarter close, the date of the firm's most recent earnings announcement).

4. Note any deviations from Part 1 definitions and justify them.

III. You are permitted to use AI to search for underlying sources (e.g., a firm's 10K) but LLMs are not permitted as the ultimate source. Citing an LLM as the source of your data may result in a grade deduction.

2. Signal Analysis (< 1,000 words):

- The table of signal values outlined above.
- Discussion of if/how you treated any missing, ambiguous, or problematic data

Step B: Combined Signal Analysis

1. Develop and justify a systematic approach to combine signals that incorporates:
 - The sign and magnitude of each coefficient from your Part 1 results
 - The statistical reliability (t-statistic)
 - The signal values that you calculated and tabulated

Hint: One simple approach to combining signals into a composite score is by ranking your candidate stocks based on a signal's value (i.e., firm with highest value =1, firm with lowest value =0, etc.). However, this aspect of the design is up to you and can be done in alternative ways.

2. Generate forced recommendations:

- The company with the highest composite score should be designated as "Buy"
- The middle company should be designated as "Neutral"
- The lowest-scoring company should be designated as "Sell"

3. Investment Recommendations (≈1,000 words):

- Detailed explanation of your signal combination methodology
- Justification for your recommendations
- Discussion of your findings from Part 1 of this project informed your approach

4. Discretionary Analysis (≈250 words):

- Assume you have submitted your report to the managing director of your firm. The director wants to stress test your recommendations by identifying critical contextual information that your quantitative models have missed
- For your three companies, discuss what contextual information the signals are missing, focusing on real-world information that are not reflected in the signals you considered. Source information from any source that you want, provided the source is credible. Clear explanation of what's missing and why it matters (e.g., pending regulatory changes, management changes or restructuring, competitive landscape shifts, geopolitical factors, etc.). Your grade depends on the intellectual depth and novelty of your answer.
- Discuss whether and why these counterpoints would lead to different portfolio choices for quant versus discretionary funds.

<< Submission Requirements >>

1. Professional report containing:
 - A one paragraph executive summary of your stock recommendation
 - All required figures and tables per the formatting guidelines. To mimic real-world expectations over attention to detail, any material deviations from the stated formatting guidelines may result in a lower grade.
 - Written answers to the analysis questions
 - A well-documented code file with your analysis
2. You are required to deliver a printed double-sided copy of the report at the beginning of the last day of class. In the header of the report, list your team member names and the name of the specific data file you were assigned.